



NOVEMBER 18, 2024

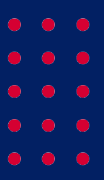
DEAN LOLA ENIOLA-ADEFESO VISIT

Undergraduate Program

Dr. Alan Zdunek
Director of Undergraduate Studies
Clinical Associate Professor



UNIVERSITY OF
ILLINOIS CHICAGO



UIC Chemical Engineering Undergraduate Program

- Program Overview
- Student Experience
- Program Improvements



Program Overview

BS Degree: 165 students Fall24

37.5% Female

31% Hispanic

20% Asian

9% International

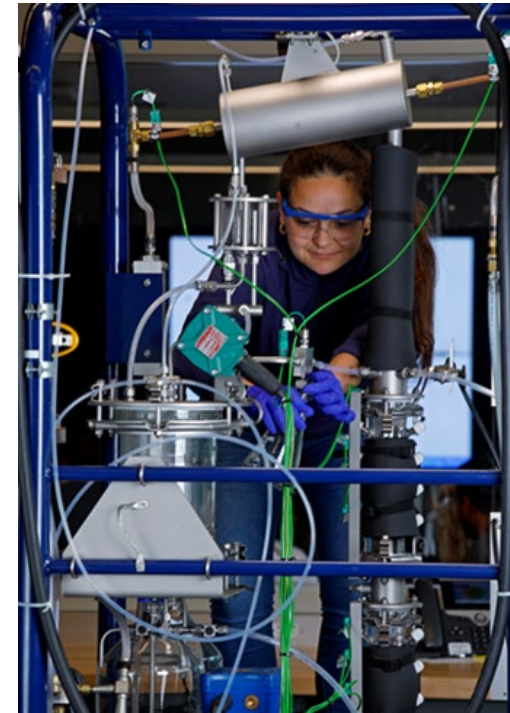
6% Black/African

5% Multi-race

Small class size:

**Private-school personal attention
at affordable public-university
tuition rates**

We are a Welcoming Community!





ChE Curriculum:

CHE 150 Climate Engineering:
General Education course
offered to Non-ChE
majors

Five concentrations
added Fall 2021

Core Courses

Thermodynamics-Based Courses

ChE 201 – Introduction to thermodynamics
ChE 301 – Chemical Engineering Thermodynamics
ChE 321 – Chemical Reaction Engineering

Process-Oriented Courses

ChE 210 – Material & Energy Balances
ChE 341 – Chemical Process Control
ChE 396 – Senior Design I (Equipment Design)
ChE 397 – Senior Design II (Design teams)

Numerical Analysis

ChE 205- Numerical Methods in ChE

Engineering Education & Development

ENGR100 -Engineering Orientation (Engineering Success Program,
Guaranteed Paid Internship Program)
ChE 499- Professional Development Seminar

Transport-Based Courses

ChE 311 – Transport Phenomena I (Fluid Mechanics)
ChE 312 – Transport Phenomena II (Heat & Mass Transfer)
ChE 313 – Transport Phenomena III (Separations)

Unit Operations Laboratory

ChE 381 – Chem Eng Lab I
ChE 382 – Chem Eng Lab II

Molecular Engineering Courses

CHE 230 Molecular Engineering in CHE
CHE 330 Polymer Science

ChE 400-level Elective
CHE 392:
Undergraduate
Research

Six Concentrations (15% of ChE students):

Biochemical Engineering	9 students
Process Automation	6
Energy & Environment	4
Nanotechnology	3
Polymers & Molecular Engineering	3
Entrepreneurship	0



Technical elective courses

+Undergraduate
research option:
(CHE 392)

4-8 students/year

CHE 410 Transport Phenomena

CHE 413 Introduction to Flow in Porous Media

CHE 421 Combustion Engineering

CHE 422 Biochemical Engineering

CHE 423 Catalytic Reaction Engineering

CHE 425 Nanotechnology for Pharmaceutical Applications

CHE 427 Entrepreneurship in Engineering

CHE 433 Process Simulation w/Aspen+

CHE 438 Computational Molecular Modeling

CHE 440 Non-Newtonian Fluids

CHE 441 Computer Applications in Chemical Engineering

CHE 451 Renewable Energy Technology

CHE 450 Air Pollution

CHE 453 Electrochemistry

CHE 454 Molecular & Macromolecular Engineering

CHE 455 Nanoscale Systems in Chemical Engineering

CHE 456. Fundamentals and Design of Microelectronics Processes

CHE 457 Colloidal and Interfacial Phenomena

CHE 494 Clean Energy & the Electric Grid

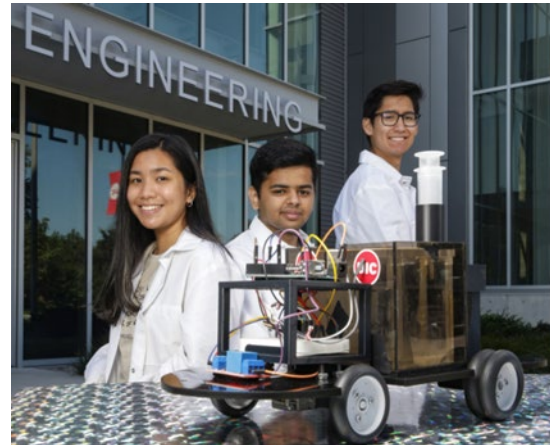
CHE 494 Biopharmaceutical Process Engineering

Student Experience



STUDENT ACTIVITIES:

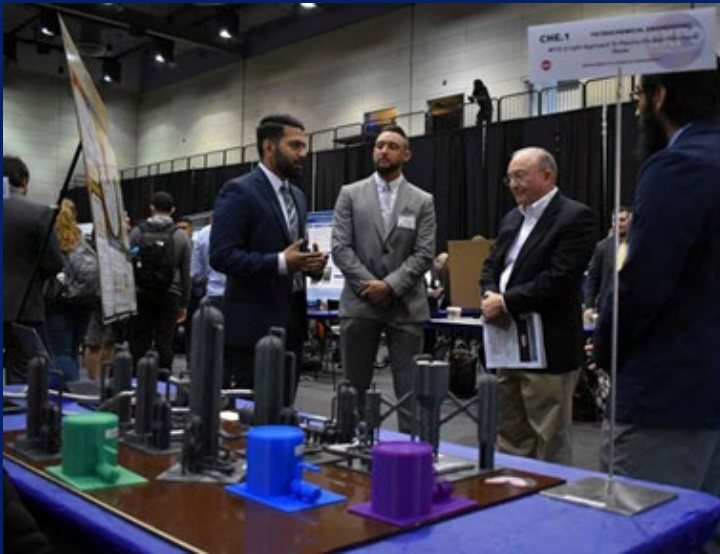
- VERY ACTIVE AIChE STUDENT CHAPTER
- ATTEND NATIONAL AIChE MEETINGS
- HOSTED NORTH-CENTRAL REGIONAL CONFERENCE 2022
- CHEME CAR, CHEME CUBE COMPETITIONS



- CO-AUTHORS ON RESEARCH PUBLICATIONS
- PARTICIPATION IN OTHER UIC STUDENT ORGANIZATIONS
- SWE AMBASSADORS
- HONORS COLLEGE
- UNDERGRADUATE MENTORS TO ENGR100



Student Experience



STUDENT/INDUSTRY/ALUMNI INTERACTION

CHE ALUMNI AND INDUSTRIAL PARTNERS PROVIDE STRONG SUPPORT TO OUR STUDENTS AND DEPARTMENT ACTIVITIES:

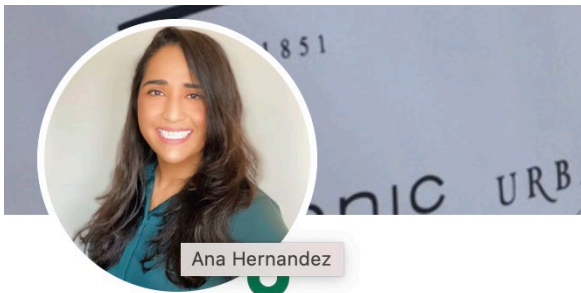
- **499 SEMINAR SPEAKERS**
- **396/397 SENIOR DESIGN MENTORS, EXPO JUDGES**
- **AICHE STUDENT CHAPTER ACTIVITIES/TOURS**
- **AICHE CHICAGO SECTION MCCORMACK AWARD**
- **AICHE CHICAGO SECTION STUDENT POSTER SESSION**
- **FEEDBACK ON CURRICULUM IMPROVEMENTS**

~750 ALUMNI IN CONTACT LIST: 2012 TO 2024 UPDATED EVERY SPRING:

UIC CHEMICAL ENGINEERING LINKEDIN GROUP

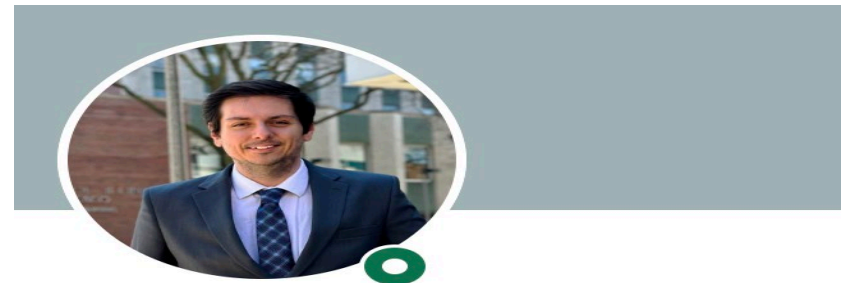


Full-time & internship placement: Our graduates are in high demand



Ana Hernandez

Ana Hernandez (She/Her) · 1st
Process Manager @L'Oréal | People Leader |



Jordan Flanigan · 1st
Semiconductor Process Engineer at Intel



Aesha Talia · 1st
Product Manager at Medline Industries, Inc.



Emir Asani · 1st
Continuous Improvement Engineer at Mars Ice Cream



Courteney Bruce-Kotey · 1st
Process Engineer at Brindley Engineering Corporation



Paul Diaz · 1st
System Manager for digital I&C and mechanical equipment at Constellation Nuclear



Program Improvement

Recent curriculum changes

PROGRAM REVISIONS:

(1) NEW COURSES

CHE 230: MOLECULAR SYSTEMS IN CHE
REQUIRED IN BS DEGREE

CHE 330: POLYMER SCIENCE

CHE 150: CLIMATE ENGINEERING FOR
GLOBAL WARMING

GENERAL EDUCATION COURSE OFFERED TO
NON-CHE MAJORS

(2) ADDED SELECTIVE OPTION TO DEGREE

CHE 330 POLYMER SCIENCE **OR**

CHEM 346 PHYSICAL CHEMISTRY II



Program Improvement

Curriculum changes in progress

(1)

ADD CHE 101:INTRODUCTION TO
SUSTAINABLE CHEMICAL ENGINEERING
TO CORE CURRICULUM (+3 HOURS)

CREATE A NEW SELECTIVE OPTION: CHE
230 MOLECULAR SYSTEMS IN CHEMICAL
ENGINEERING OR CME 260 PROPERTIES
OF MATERIALS (-3 HOURS)

(2)

REMOVE CHEM 234 CHEMICAL SYNTHESIS
(ORG CHEM II)FROM MATH/SCIENCE
REQUIREMENT (-3 HOURS)

ADD SECOND TECHNICAL ELECTIVE TO
CORE REQUIREMENT (+3 HOURS)



Program Improvement: Chemical Engineering & Data Science

CHEMICAL ENGINEERING AND DATA SCIENCE

ADD DATA SCIENCE TO OUR CURRICULUM

(1)

**BS IN CHE WITH DATA SCIENCE
CONCENTRATION**

**(UIUC IS IMPLEMENTING A SIMILAR
SPECIALIZATION)**

(2)

**BS IN DATA SCIENCE WITH CHE
CONCENTRATION**

**(UIC COMPUTER SCIENCE DEPARTMENT HAS
SEVERAL PROGRAMS WITH OTHER MAJORS)**



Program Improvement: Recruitment Efforts

**EMPHASIZING CHEMICAL ENGINEERS' ROLE IN
SUSTAINABILITY AT OUR RECRUITMENT EVENTS:**

UIC OPEN HOUSE, IGNITE EVENT

HIGH SCHOOL VISITS

COMMUNITY COLLEGE VISITS

WOMEN IN ENGINEERING SCHOLARSHIP BRUNCH

WIE SUMMER CAMP

PERSONALIZED COMMUNICATION TO APPLICANTS →ITE!

**CHEMICAL ENGINEERING: SHAPING A SUSTAINABLE
FUTURE**

DISCOVER HOW CHEMICAL ENGINEERS ARE TRANSFORMING OUR FUTURE WITH SUSTAINABLE SOLUTIONS, FROM RENEWABLE ENERGY, CLEAN WATER AND ENVIRONMENTAL TECHNOLOGIES TO LIFE-SAVING DRUGS, SUSTAINABLE FOOD AND ECO-FRIENDLY PRODUCTS. SEE HOW YOU CAN JOIN THIS EXCITING FIELD AND MAKE A REAL DIFFERENCE IN THE WORLD!

